PARAMETRIC MODERN CABINET SYSTEM

Open Shelf 360 x 500

Assembly guide - Open mini bookshelf 360 x 500 x 230 mm, 1 shelf, back panel - tool-free dovetail



Key facts

Size (W x H x D)	360 x 500 x 230 mm
Printer	Bambu Lab P2S
Build plates	15
Print time / filament	68 h / 2.7 kg
Filament	PETG, 3 walls, 15 % gyroid, 0.2 mm

> Safety: secure the cabinet to the wall (two holes in the top panel). The calculations are engineering estimates for printed PETG, not a test report.

Hardware you need

- 0 x M4 hex nut for the built-in handle posts
- 2 x Wall screw + plug or anti-tip strap for the 2 anchor holes in the top panel (wall dependent)

Tools: rubber mallet, glue (optional); drill for the wall anchor

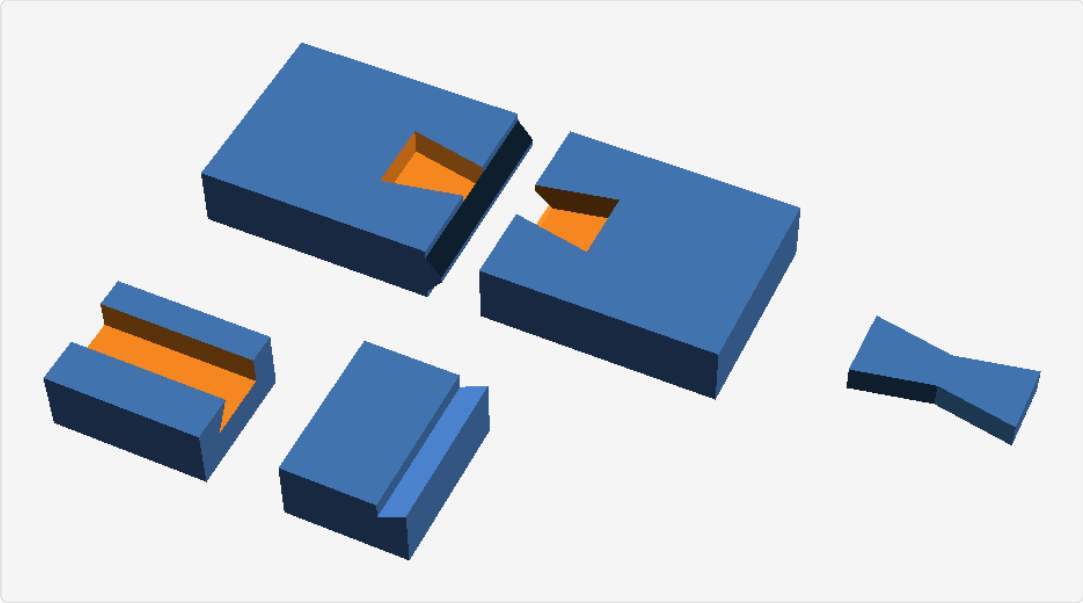
Print plan

Plate	Objects on the plate
1	sideL0[0,1], keys x5
2	sideR0[0,1], keys x5
3	sideL0[0,0], keys x5, keys x5
4	sideR0[0,0], keys x4
5	hp0[0,0]
6	hp1[0,0]
7	hp2[0,0]
8	back0[0,0]
9	back0[0,1]
10	back0[1,0]
11	back0[1,1]
12	hp0[1,0]
13	hp1[1,0]
14	hp2[1,0]
15	sideL0[0,2], sideR0[0,2]

Names: sideL / sideR = side panels, hp0 = bottom, hp1.. = shelves, last hp = top, back = back panel, door = door, [i,j] = tile column / row (tiles are joined by tongue-and-groove and bow-tie keys), keys = bow-tie keys, leaf hinges = print-in-place hinges.

0 Print and prepare

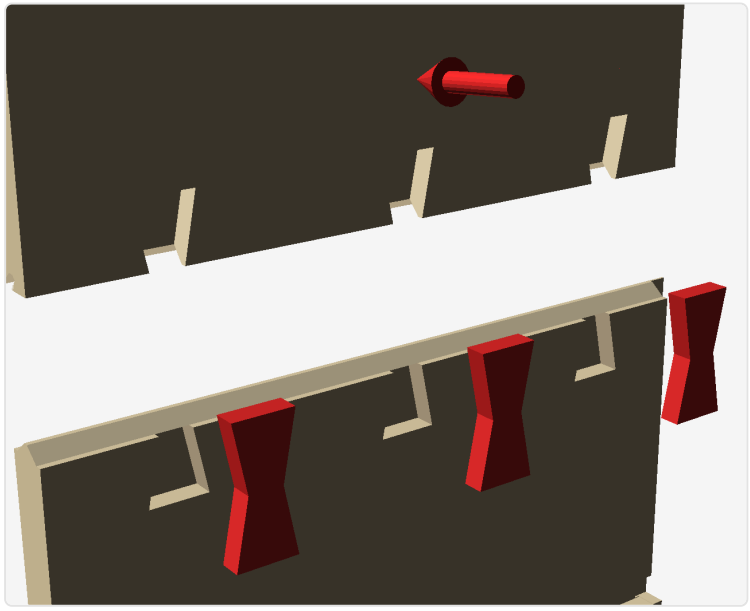
Print every plate with **PETG** (3 wall loops, 15 % gyroid, 0.2 mm). Print the **fit-test coupon** (`fit-test.stl`) first: the dovetail block must slide by hand with light friction, the hinge pin must turn freely and the bow-tie key must press in with your thumb. If it is too tight / loose, change `fit` (0.25 default) or `hinge_fit` in the Customizer and generate the plates again.



Fit-test coupon: print it first (dovetail, bow-tie key, hinge, M3 boss).

1 Join the tiles of every big panel

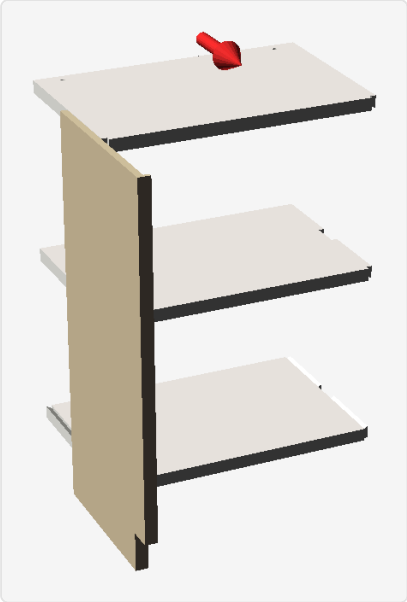
Large panels are printed as bed-sized tiles. Lay the tiles of one panel next to each other, **tongue into groove**, face-down on the table. Press the **bow-tie keys** into the pockets on the hidden face (the face that was *up* on the printer: inner face of the sides, underside of the shelves, back of the doors). Keys sit 0.4 mm below the surface; a drop of cyanoacrylate makes the seam permanent. The back panel has plain tongue-and-groove seams (add a few drops of glue or a strip of tape on the rear).



Tongue into groove. Press the bow-tie keys (red) into the pockets of the hidden face.

3 Slide the shelves into the left side

Stand the left side on the table (inner face up). Push **every horizontal panel from the rear** into its dovetail groove until it stops against the blind front end. The tenon blocks lock the panel against the side - it cannot be pulled out sideways.



Slide every horizontal panel in from the rear (arrow).

4 Slide the right side on

Place the right side in front of the shelf ends and slide it **backwards** over all tenons at once until its grooves stop. Use a rubber mallet if the fit is tight.



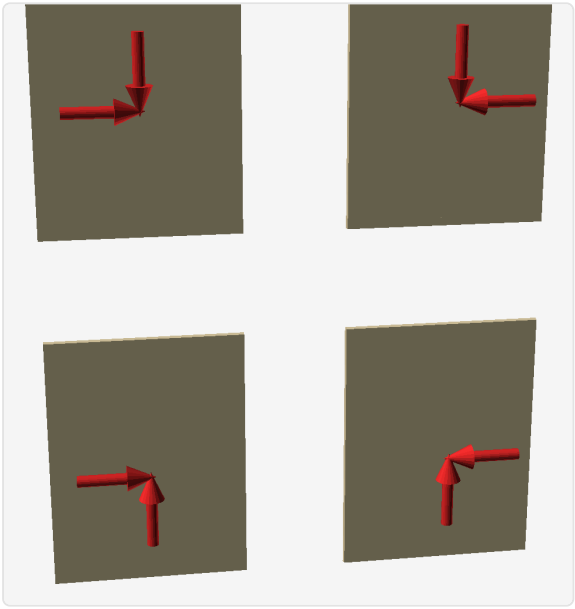
Slide the right side on from the front, over all tenons at once (arrow).

5 Back panel

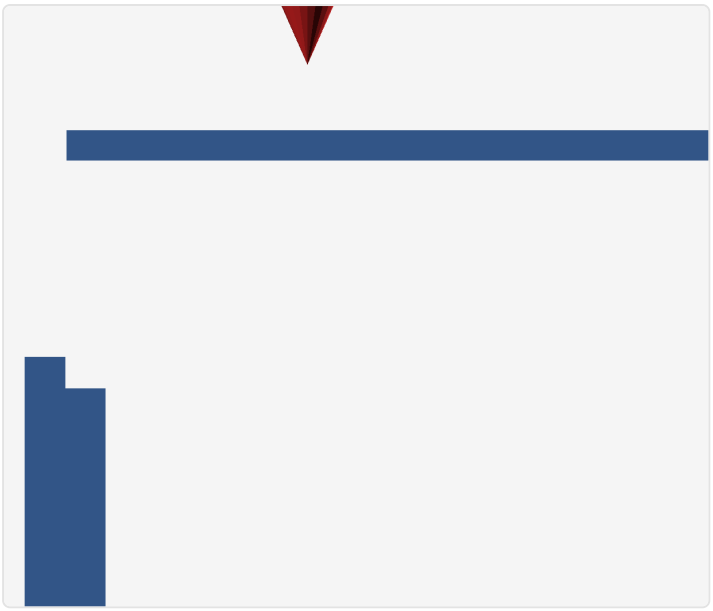
Slide the back panel into the rabbet from the rear. It also locks all shelves against sliding out backwards. Friction fit - add a few drops of glue or tape if you want it permanent.



5 Overview: the back panel is slid in from the rear (arrow).



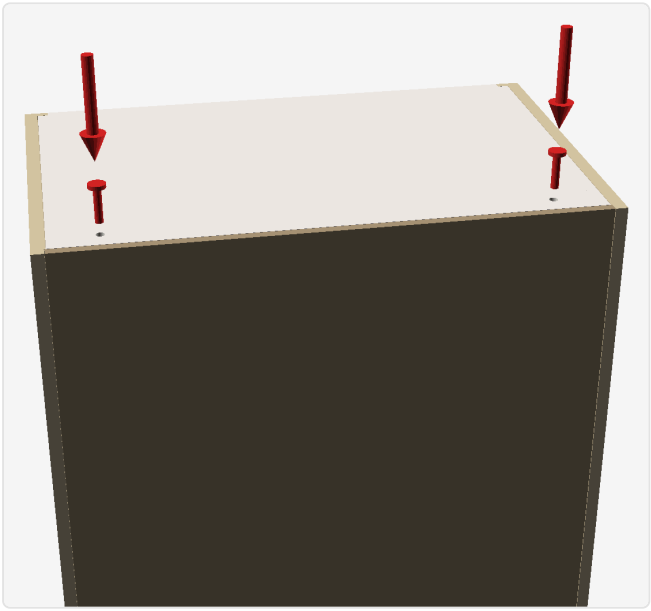
5a Lay the tiles of the back panel face-down in the arrangement shown and push them together (arrows): tongue into groove. A few drops of glue or a strip of tape on the rear make the seams permanent.



5b Section from above: the finished back panel slides from the rear (arrow) into the rabbet at the rear edge of both side panels and ends flush with them.

9 Secure the cabinet to the wall

Mandatory. The two holes in the top panel take wall brackets or an anti-tip strap. Free-standing, an empty cabinet of this kind tips forward under a small load on its front edge (see *tip-over* in the verification summary below).



9 Wall anchor: the two holes (red = screws) in the top panel near the rear edge take the wall screws / anti-tip strap (arrows).



Finished cabinet, door opened.

Appendix - what was verified

Every assembly step above was simulated on the real parts (swept-path collision tests); strength values are conservative estimates.

Assembly and statics

	Check	Result
OK	S1 3 horizontal panels slide from the rear into the left side (tool-free)	worst intersection 0.00 mm3 over 16 positions each
OK	S2 right side slides from the front over all tenons simultaneously	worst 0.00 mm3
OK	S3 back panel slides into the rabbet from the rear	worst 0.00 mm3
OK	L1a shelves cannot be pushed out backwards once the back panel is on	
OK	L1b shelves cannot slide forwards (blind groove stop)	
OK	L2 sides cannot be pulled apart (half-dovetail locks every shelf)	
-	mass model	cabinet 2.7 kg (solid PETG 10.2 kg, fill factor 0.27 calibrated on the slicer weight); centre of mass x=180 y=127 z=265
-	tip-over (front edge load)	load that tips the empty cabinet forward: 9 kg; loaded 23 kg: 66 kg (lever = toe-kick 30 mm)
OK	stability class: wall anchor REQUIRED	top panel carries the 2 anchor holes (wall_anchor = true)
OK	centre of mass is centred between the sides	offset +0 mm
OK	shelf rating: 15 kg evenly distributed per shelf (stress and deflection <= L/200)	span 328 mm, shelf 16 mm thick
OK	shelf seam (bow-tie keys as tension ties) carries the bending moment	capacity 36.0 N*m vs demand 6.03 N*m (15 kg), safety 6.0
OK	dovetail bearing pressure on the sloped flank	0.15 MPa (allow 3.0) per shelf end, 2 blocks
OK	tenon neck shear (across layers)	0.044 MPa (allow 1.33)

Printability

	Check	Result
OK	all 16 tiles fit the bed (256x256)	
OK	all tiles lie flat on the bed (zmin = 0)	
OK	every tile is ONE connected body	
OK	no large unsupported overhangs / long bridges (>10.5 mm)	
OK	tile seams (11): tongues sit in grooves without interference	
OK	assembly (closed): no part-part interference (8 pairs)	
OK	shelf 1 <-> left side clearance	gap=0.150 (design 0.15)
OK	back panel <-> side clearance	gap=0.200 (design 0.1)
OK	hardware part 'key' prints flat, one body, no steep overhang	z0=0.00 bodies=1 overhangs=[] size=[40.0, 18.0, 6.6]